

PRANJAL MONDAL

+91-8653928123 | mondalpranjal28@gmail.com

CAREER OBJECTIVE

Industrious and motivated Computer Science undergraduate with a zeal for innovation and a strong foundation in software development, algorithms, and emerging technologies. Eager to contribute to impactful, real-world engineering challenges while continuously expanding technical expertise. Passionate about building efficient, scalable systems and leveraging AI/ML to solve complex problems. Seeking opportunities to apply academic knowledge and project experience in a dynamic professional environment.

EDUCATION

Bachelor of Engineering - Computer Science & Engineering Jadavpur University, Kolkata	Expected: 2027 CGPA: 7.6/10
Higher Secondary Education (Class XII) Ghatal Vidyasagar High School, West Bengal	Batch of 2023 WBJEE Rank: 144

RELEVANT COURSES

- Data Structures
- Database Management Systems
- Software Design Methodology
- Algorithm Design
- Computer Vision

TECHNICAL SKILLS

Programming Languages	Java (Web Development), JavaScript (Node.js), C, C++, Python
Web & API Development	RESTful API Integration, Web Application Development, Client-Server Architecture
Software Engineering	Program Development, Software Design Methodology, Object-Oriented Design
AI/ML	Machine Learning fundamentals, Computer Vision, Neural Networks, Data preprocessing & model evaluation
Databases	Relational Database Design, SQL, Database Management Systems (DBMS)
Tools & Platforms	Git, VS Code, Linux/Unix, Command-line tools

PROJECTS

Multi-Client Chat Application

Technologies: Node.js, Socket.io, Express, JavaScript, REST API, Event-Driven Architecture

- Designed and implemented a full-stack multi-client chat application using Node.js with separate, modular client and server programs supporting simultaneous real-time communication between multiple users.
- Built a concurrent server capable of broadcasting text messages as well as image/video media to all connected clients, with graceful handling of clients joining and leaving dynamically.
- Implemented one-to-one private messaging and chatroom features, enabling both group and direct communication channels within the same application.
- Developed an AI-powered troll detection module that invokes an LLM (Gemini/ChatGPT) API to generate context-aware, empathetic responses when trolling is detected in chatrooms, avoiding stern warnings in favour of soothing, constructive messages.
- Applied design patterns including Observer (event-driven pub/sub messaging), Middleware Chain (Express request pipeline), and Singleton (shared server state); analysed resulting protocol layers and documented findings in a detailed experimental report.

HOBBIES & INTERESTS

- **Learning about the Stock Market:** Enthusiastic about studying financial markets, tracking economic trends, and analyzing company fundamentals (such as balance sheets and earnings reports). Actively exploring investment strategies, risk management, and the macroeconomic factors that drive market dynamics to build a solid foundation in wealth management.